DATA MINING PROJECT PROPOSAL

Predicting Loan Default Risk Using LendingClub Data

Course: Data Mining
Dataset: LendingClub Loan Data
Type: Binary Classification
Records: Accepted Loans Dataset

1. PROBLEM STATEMENT & MOTIVATION

Consumer lending carries substantial financial risk. While LendingClub assigns letter grades (A–G) to each loan, these grades do not fully capture the true default probability at the individual loan level. This project asks: Can a machine learning model predict whether an accepted loan will be fully repaid or charged off — and can it outperform LendingClub's own grading system as a risk signal?

This is a high-value, real-world problem in the credit risk domain. Accurate default prediction enables lenders to minimize losses while extending credit fairly and efficiently.

2. RESEARCH OBJECTIVE

Build a binary classification model that predicts loan outcome (Fully Paid vs. Charged Off) using borrower and loan characteristics available at the time of issuance — with no post-origination leakage.

Secondary objective: Evaluate whether the model adds predictive value beyond LendingClub's existing grade/sub-grade system, and identify the key drivers of default risk using explainability methods.

3. DATASET & TARGET VARIABLE

Source: Two CSV files from LendingClub — Accepted Loans and Rejected Applications.

Primary analysis dataset: Accepted Loans, filtered to loan_status ∈ {Fully Paid, Charged Off}. The Rejected file will be used for exploratory comparison only, as it contains far fewer features.

Target variable: loan_status (binary: 0 = Fully Paid, 1 = Charged Off)

Key predictive features (measured at origination):

- Credit profile: fico_range_low/high, inq_last_6mths, delinq_2yrs, revol_util, pub_rec
- Loan characteristics: loan_amnt, int_rate, term, grade, sub_grade, purpose
- Borrower profile: annual_inc, dti, emp_length, home_ownership, addr_state
- Engineered features: loan-to-income ratio, credit age, FICO drop (original vs. last pull)

Note: Post-origination columns (total_pymnt, out_prncp, recoveries, etc.) will be excluded to prevent data leakage.

4. METHODOLOGY

Phase	Description
EDA	Default rate by grade, state, income; FICO and DTI distributions; correlation analysis
Preprocessing	Handle missing values, encode categoricals, normalize numerics, engineer ratio features
Imbalance Handling	Address class imbalance (~20% default rate) using SMOTE and/or class_weight adjustment
Modeling	Logistic Regression (baseline), Random Forest, XGBoost — with cross-validation

Evaluation	ROC-AUC, Precision-Recall curve, F1 score — accuracy alone is misleading on imbalanced data
Explainability	SHAP values to identify top default predictors; compare model vs. LendingClub grade as risk signal

5. EXPECTED CONTRIBUTIONS

- A well-validated default prediction model with interpretable feature importance rankings
- Analysis of whether a data-driven model can add value beyond LendingClub's own grading system
- Insights into which borrower characteristics most strongly signal default risk
- Demonstration of proper ML pipeline practices: leakage prevention, imbalance handling, and model explainability

6. TOOLS & TECHNOLOGIES

Python (pandas, scikit-learn, XGBoost, imbalanced-learn, SHAP, matplotlib, seaborn)